Research & Knowledge Discovery Base

A multi-domain reference document spanning technology, science, humanities, and social sciences. Intended as an external knowledge source for retrieval-augmented generation (RAG) pipelines built for research and knowledge discovery use cases.

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1. Networking Fundamentals

1.1 The OSI Model

The OSI (Open Systems Interconnection) model divides network communication into seven layers. From bottom to top these are: Layer 1 Physical (raw bit transmission over cables, fiber, or radio), Layer 2 Data Link (framing and MAC addressing, e.g. Ethernet and switches), Layer 3 Network (logical addressing and routing, e.g. IP and routers), Layer 4 Transport (end-to-end delivery, e.g. TCP and UDP), Layer 5 Session (managing sessions between applications), Layer 6 Presentation (data translation, encryption, compression), and Layer 7 Application (the interface users and applications interact with, e.g. HTTP, DNS, FTP). Each layer only communicates with the layers directly above and below it, which allows protocols to be developed and troubleshot independently.

1.2 TCP vs UDP

TCP (Transmission Control Protocol) is connection-oriented: it establishes a session using a three-way handshake (SYN, SYN-ACK, ACK), guarantees ordered and reliable delivery through acknowledgments and retransmission, and includes flow and congestion control. This makes TCP suitable for web browsing, email, and file transfer. UDP (User Datagram Protocol) is connectionless: it sends datagrams with no handshake, no guarantee of delivery or ordering, and minimal overhead. UDP is preferred for real-time applications such as VoIP, video streaming, and DNS lookups, where low latency matters more than perfect reliability.

1.3 IP Addressing and Subnetting

IPv4 addresses are 32-bit values written as four decimal octets (e.g. 192.168.1.10). A subnet mask (e.g. 255.255.255.0, or /24 in CIDR notation) separates the network portion from the host portion of an address. Private IP ranges reserved by RFC 1918 are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16; these are not routable on the public internet and require Network Address Translation (NAT) to communicate externally. IPv6 uses 128-bit addresses written in hexadecimal (e.g. 2001:0db8::1) and was designed to solve IPv4 address exhaustion.

1.4 Key Networking Devices

- **Router:** forwards packets between different networks based on IP addresses; operates at Layer 3.
- **Switch:** forwards frames within a local network based on MAC addresses; operates at Layer 2.
- **Firewall:** filters traffic based on rules (IP, port, protocol, or application behavior) to enforce security policy.
- **Load Balancer:** distributes incoming traffic across multiple servers to improve availability and performance.
- **Proxy Server:** intermediates requests between clients and servers, often for caching, filtering, or anonymization.

1.5 DNS and DHCP

DNS (Domain Name System) resolves human-readable domain names (e.g. example.com) into IP addresses using a distributed, hierarchical database of root servers, TLD servers, and authoritative name

servers. DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, default gateways, and DNS servers to devices on a network via a four-step process commonly abbreviated as DORA: Discover, Offer, Request, Acknowledge.

2. Cybersecurity & SOC/NOC Operations

2.1 SOC vs NOC

A Security Operations Center (SOC) is a team responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity incidents. A Network Operations Center (NOC) is responsible for monitoring network performance, availability, and infrastructure health (bandwidth, uptime, latency, outages). The SOC focuses on security threats such as intrusions, malware, and data breaches, while the NOC focuses on operational continuity such as link failures, hardware faults, and performance degradation. In many organizations the two teams share tooling and escalate incidents to each other, since a security event (e.g. a DDoS attack) often shows up first as a NOC-visible performance issue.

2.2 The SOC Analyst Tiered Model

- **Tier 1 (Triage):** monitors alerts from SIEM/EDR tools, performs initial validation, and escalates true positives.
- **Tier 2 (Incident Responder):** investigates escalated alerts, performs deeper analysis, and contains active threats.
- **Tier 3 (Threat Hunter / Senior Analyst):** proactively searches for hidden threats, reverse engineers malware, and refines detection rules.
- **SOC Manager:** oversees operations, reporting, and coordination with other teams (legal, PR, executives) during major incidents.

2.3 Core Security Tools

A SIEM (Security Information and Event Management) system aggregates and correlates log data from across an organization to detect suspicious patterns and generate alerts; common examples include Splunk, IBM QRadar, and Microsoft Sentinel. An EDR (Endpoint Detection and Response) tool monitors individual devices for malicious behavior and can isolate or remediate compromised endpoints. A SOAR (Security Orchestration, Automation, and Response) platform automates repetitive incident-response workflows (e.g. blocking an IP or disabling a user account) by chaining playbooks across multiple tools, which is the main entry point for AI-driven SOC/NOC automation. An IDS/IPS (Intrusion Detection/Prevention System) inspects network traffic for known attack signatures or anomalies, either alerting (IDS) or actively blocking (IPS).

2.4 The Incident Response Lifecycle

The widely used NIST incident response lifecycle has four phases: (1) Preparation, building policies, tools, and training before an incident occurs; (2) Detection and Analysis, identifying and validating that an incident has occurred and determining its scope; (3) Containment, Eradication, and Recovery, isolating affected systems, removing the threat, and restoring normal operations; and (4) Post-Incident Activity, documenting lessons learned and updating defenses to prevent recurrence.

2.5 Common Attack Types

- **Phishing:** deceptive messages tricking users into revealing credentials or installing malware.
- **Ransomware:** malware that encrypts victim data and demands payment for the decryption key.

- **DDoS (Distributed Denial of Service):** overwhelming a target with traffic from many sources to disrupt availability.
- **Man-in-the-Middle (MITM):** intercepting and potentially altering communication between two parties without their knowledge.
- **SQL Injection:** inserting malicious SQL through unvalidated input fields to manipulate or exfiltrate database contents.
- **Zero-Day Exploit:** an attack that targets a previously unknown vulnerability before a patch exists.

2.6 Key Frameworks

The MITRE ATT&CK framework is a curated knowledge base of real-world adversary tactics and techniques (e.g. Initial Access, Persistence, Privilege Escalation, Exfiltration), widely used to map detections and threat-hunting coverage. The Cyber Kill Chain, developed by Lockheed Martin, models an attack as a sequence of stages: Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command and Control, and Actions on Objectives; defenders aim to break the chain at the earliest possible stage.

3. Artificial Intelligence & Large Language Models

3.1 What Is a Large Language Model

A Large Language Model (LLM) is a neural network, typically based on the Transformer architecture, trained on massive amounts of text to predict the next token in a sequence. Through this next-token prediction objective at scale, LLMs learn statistical patterns of grammar, facts, reasoning structures, and style, which allows them to generate coherent text, answer questions, summarize content, and follow instructions after further fine-tuning.

3.2 The Transformer Architecture

Introduced in the 2017 paper 'Attention Is All You Need', the Transformer relies on a self-attention mechanism that lets each token in a sequence weigh the relevance of every other token when building its representation, rather than processing tokens strictly in order as older RNN-based models did. This enables much greater parallelization during training and better handling of long-range dependencies in text. Key components include multi-head self-attention, position-wise feed-forward layers, positional encodings (since attention itself has no notion of token order), and residual connections with layer normalization.

3.3 Fine-Tuning vs PEFT

Full fine-tuning updates all of a pretrained model's parameters on a task-specific dataset, which is effective but computationally expensive and requires storing a full copy of the model per task. Parameter-Efficient Fine-Tuning (PEFT) methods instead freeze most of the original model and train only a small number of additional parameters, dramatically reducing compute and storage costs while retaining most of the performance benefit. Common PEFT techniques include LoRA, adapters, and prompt/prefix tuning.

3.4 LoRA and QLoRA

LoRA (Low-Rank Adaptation) freezes the pretrained weight matrices of a model and injects small trainable low-rank matrices into each layer (typically into the attention projections). Instead of updating a full weight matrix W, LoRA learns two much smaller matrices A and B such that the update is approximated as W + BA, where the rank of BA is far lower than that of W. This cuts the number of trainable parameters by orders of magnitude while matching much of full fine-tuning's performance. QLoRA extends this by first quantizing the frozen base model to 4-bit precision (using a technique called NF4) to shrink its memory footprint, then applying LoRA adapters on top in higher precision, allowing large models to be fine-tuned on a single consumer or prosumer GPU.

3.5 Retrieval-Augmented Generation (RAG)

RAG combines an LLM with an external knowledge retrieval step so the model can ground its answers in up-to-date or domain-specific information it was not trained on. A typical pipeline: (1) source documents are split into chunks; (2) each chunk is converted into a vector embedding using an embedding model and stored in a vector database (e.g. FAISS, Pinecone, Chroma, Weaviate); (3) when a user asks a question, the query is embedded and the most similar chunks are retrieved via nearest-neighbor search; (4) the retrieved chunks are inserted into the LLM's prompt as context, and the model generates an answer grounded in that context. RAG reduces hallucination, allows knowledge updates without retraining the

model, and can cite sources for transparency.

3.6 Embeddings and Vector Search

An embedding is a dense numerical vector representation of text (or images, audio, etc.) such that semantically similar inputs are located close together in vector space. Similarity between embeddings is commonly measured with cosine similarity or dot product. Approximate Nearest Neighbor (ANN) algorithms such as HNSW (Hierarchical Navigable Small World) allow vector databases to search millions of embeddings for the closest matches in milliseconds, trading a small amount of accuracy for large gains in speed.

3.7 Prompt Engineering and Agents

Prompt engineering is the practice of crafting inputs to an LLM to reliably elicit a desired output, using techniques such as few-shot examples, chain-of-thought prompting (asking the model to reason step by step), and structured output formatting. An AI agent extends an LLM with the ability to take actions: it can call external tools or APIs, observe the results, and decide on a next step in a loop, which allows it to perform multi-step tasks such as querying a database, browsing the web, or executing code rather than just generating text.

4. Cloud Computing Basics

4.1 Service Models

- **IaaS (Infrastructure as a Service):** provides raw virtualized compute, storage, and networking (e.g. AWS EC2, Azure VMs); the customer manages the OS and above.
- **PaaS (Platform as a Service):** provides a managed runtime/platform for deploying applications (e.g. AWS Elastic Beanstalk, Google App Engine) without managing servers.
- **SaaS (Software as a Service):** delivers a complete, ready-to-use application over the internet (e.g. Gmail, Salesforce), with the provider managing everything underneath.

4.2 Deployment Models

A public cloud is owned and operated by a third-party provider (e.g. AWS, Azure, GCP) and shared across multiple customers over the internet. A private cloud is dedicated to a single organization, either on-premises or hosted, offering more control at higher cost. A hybrid cloud combines public and private environments, allowing workloads to move between them, which is common for organizations with regulatory or latency constraints on certain data.

4.3 Containers and Orchestration

A container packages an application together with its dependencies into a single lightweight, portable unit that shares the host operating system's kernel, in contrast to a virtual machine which virtualizes an entire OS. Docker is the most widely used container runtime and image format. Kubernetes is an orchestration platform that automates the deployment, scaling, networking, and self-healing of containerized applications across a cluster of machines, using concepts such as Pods (the smallest deployable unit), Deployments (manage replica sets of Pods), and Services (stable networking endpoints for Pods).

4.4 Scalability Concepts

Vertical scaling (scaling up) increases the resources (CPU, RAM) of an existing server, which is simple but has a hard ceiling and typically requires downtime. Horizontal scaling (scaling out) adds more server instances and distributes load across them, offering higher availability and near-limitless growth but requiring the application to be designed to run across multiple nodes (e.g. stateless services, shared session storage). Auto-scaling automatically adds or removes instances based on real-time demand metrics such as CPU utilization or request rate.

4.5 CI/CD

Continuous Integration (CI) is the practice of frequently merging code changes into a shared repository, with automated builds and tests run on every change to catch integration issues early. Continuous Delivery/Deployment (CD) extends this by automatically packaging and releasing validated code to staging or production environments. A typical CI/CD pipeline (e.g. using GitHub Actions, GitLab CI, or Jenkins) includes stages for build, automated testing, security/quality scanning, and deployment, reducing manual effort and the risk of human error in releases.

5. Space & Astronomy

5.1 The Solar System

The Solar System consists of the Sun and everything gravitationally bound to it: eight planets, their moons, dwarf planets (such as Pluto, Ceres, and Eris), asteroids, comets, and interplanetary dust. The four inner planets (Mercury, Venus, Earth, Mars) are small and rocky (terrestrial), while the four outer planets (Jupiter, Saturn, Uranus, Neptune) are much larger gas or ice giants. The asteroid belt between Mars and Jupiter separates these two groups. The Sun contains more than 99.8% of the Solar System's total mass and generates energy through nuclear fusion, converting hydrogen into helium in its core.

5.2 Stars and Stellar Life Cycles

Stars form from collapsing clouds of gas and dust called nebulae. A star's mass determines its life cycle: low-mass stars like the Sun eventually expand into red giants, shed their outer layers as a planetary nebula, and leave behind a dense core called a white dwarf. High-mass stars burn through their fuel much faster, then end in a supernova explosion, leaving behind either a neutron star or, if massive enough, a black hole. The color and temperature of a star are closely related: blue stars are hottest, red stars are coolest, a relationship plotted on the Hertzsprung-Russell diagram.

5.3 Galaxies and the Universe

A galaxy is a massive, gravitationally bound system of stars, stellar remnants, gas, dust, and dark matter. The Milky Way, our home galaxy, is a barred spiral galaxy containing an estimated 100 to 400 billion stars. The observable universe contains hundreds of billions of galaxies. The Big Bang theory describes the universe as originating from an extremely hot, dense state approximately 13.8 billion years ago and has been expanding ever since; this expansion is accelerating due to a poorly understood force termed dark energy, which along with dark matter is estimated to make up around 95% of the universe's total mass-energy content.

5.4 Space Exploration Milestones

- **1957:** Sputnik 1, the first artificial satellite, launched by the Soviet Union.
- **1961:** Yuri Gagarin becomes the first human in space aboard Vostok 1.
- **1969:** Apollo 11 lands the first humans, Neil Armstrong and Buzz Aldrin, on the Moon.
- **1990:** The Hubble Space Telescope is launched, transforming observational astronomy.
- **2021:** The James Webb Space Telescope launches, designed to observe in infrared and see further back in cosmic time than Hubble.

6. Human Biology & Health

6.1 Cell Structure and Function

The cell is the basic structural and functional unit of life. Human cells are eukaryotic, containing a nucleus that houses DNA organized into chromosomes. Key organelles include mitochondria, which generate ATP (the cell's energy currency) through cellular respiration; the endoplasmic reticulum and Golgi apparatus, which process and package proteins; and the cell membrane, a selectively permeable phospholipid bilayer that regulates what enters and exits the cell. The human body contains an estimated 30 trillion cells across roughly 200 distinct cell types.

6.2 The Circulatory and Respiratory Systems

The circulatory system, driven by the heart, transports oxygen, nutrients, hormones, and waste products throughout the body via blood vessels: arteries carry oxygenated blood away from the heart, veins return deoxygenated blood, and capillaries enable exchange at the tissue level. The respiratory system, centered on the lungs, brings oxygen into the bloodstream and removes carbon dioxide; gas exchange occurs across tiny air sacs called alveoli. The heart and lungs work together in the pulmonary and systemic circuits to keep tissues supplied with oxygen.

6.3 The Immune System

The immune system defends the body against pathogens through two main branches. Innate immunity is the body's fast, non-specific first line of defense, including physical barriers (skin, mucus) and cells like macrophages and neutrophils that attack anything foreign. Adaptive immunity is slower to activate but highly specific: B cells produce antibodies tailored to particular pathogens, and T cells directly kill infected cells or help coordinate the response. Memory cells formed during adaptive immunity are the basis for both long-term immunity after infection and the effectiveness of vaccines.

6.4 DNA, Genes, and Heredity

DNA (deoxyribonucleic acid) is a double-helix molecule made of four nucleotide bases (adenine, thymine, guanine, cytosine) that encodes the instructions for building and operating an organism. A gene is a segment of DNA that typically codes for a specific protein. Humans have roughly 20,000 to 21,000 protein-coding genes organized across 23 pairs of chromosomes. Traits are inherited from both parents according to patterns first described by Gregor Mendel, including dominant and recessive alleles, though most human traits are influenced by multiple genes and environmental factors.

6.5 Nutrition Basics

The body requires macronutrients (carbohydrates, proteins, and fats) for energy and structural needs, and micronutrients (vitamins and minerals) in smaller amounts to support metabolic processes. Carbohydrates are the body's primary energy source; proteins are broken down into amino acids used to build and repair tissue; fats support cell membrane structure, hormone production, and long-term energy storage. A balanced diet combined with regular physical activity is broadly associated with reduced risk of chronic conditions such as cardiovascular disease and type 2 diabetes.

7. Environmental Science & Climate

7.1 The Greenhouse Effect and Climate Change

The greenhouse effect is a natural process in which certain atmospheric gases (primarily carbon dioxide, methane, and water vapor) trap heat radiated from Earth's surface, keeping the planet warm enough to support life. Since the Industrial Revolution, human activity, particularly the burning of fossil fuels and deforestation, has sharply increased atmospheric concentrations of these greenhouse gases, intensifying the effect and driving long-term global warming. This is linked to rising sea levels from melting ice sheets and thermal expansion of seawater, more frequent extreme weather events, and shifting ecosystems and growing seasons.

7.2 Renewable Energy Sources

- **Solar power:** converts sunlight into electricity via photovoltaic cells, or into heat via solar thermal systems.
- **Wind power:** uses turbines to convert the kinetic energy of moving air into electricity.
- **Hydropower:** generates electricity from the energy of flowing or falling water, typically via dams.
- **Geothermal energy:** taps heat from beneath the Earth's surface for electricity generation or direct heating.
- **Biomass energy:** derives energy from organic material such as wood, crop residue, or waste.

7.3 Ecosystems and Biodiversity

An ecosystem is a community of living organisms interacting with each other and their physical environment, connected through energy flow (e.g. food chains and food webs) and nutrient cycles (such as the carbon, nitrogen, and water cycles). Biodiversity refers to the variety of life at genetic, species, and ecosystem levels, and is widely considered important for ecosystem resilience and services such as pollination, water purification, and climate regulation. Habitat destruction, pollution, climate change, and invasive species are among the leading drivers of biodiversity loss globally.

7.4 The Water Cycle

The water cycle describes the continuous movement of water through the atmosphere, land, and oceans via evaporation (liquid water turning to vapor), transpiration (water release from plants), condensation (vapor forming clouds), precipitation (rain, snow, or hail falling to Earth), and runoff and infiltration (water flowing over or soaking into the ground, eventually returning to rivers, lakes, and oceans). This cycle is powered primarily by solar energy and is essential for redistributing fresh water across the planet.

8. World History

8.1 Ancient Civilizations

Some of the earliest known civilizations arose along fertile river valleys: Mesopotamia (Tigris and Euphrates rivers), Ancient Egypt (the Nile), the Indus Valley Civilization (the Indus river, in present-day Pakistan and northwest India), and Ancient China (the Yellow River). These civilizations independently developed writing systems, organized agriculture, centralized governments, and monumental architecture. Mesopotamia is often called the 'cradle of civilization' due to early developments such as cuneiform writing, the wheel, and the earliest known legal codes, including the Code of Hammurabi.

8.2 The Industrial Revolution

The Industrial Revolution began in Britain in the mid-18th century and spread through Europe and North America over the following century. It marked a shift from agrarian, hand-production economies to ones dominated by machine manufacturing, powered initially by the steam engine and later by electricity and the internal combustion engine. Key developments included mechanized textile production, the growth of factories and urban centers, expansion of railways, and significant increases in productivity and population, alongside harsh working conditions that eventually spurred labor rights movements.

8.3 The World Wars

World War I (1914-1918) was triggered by the assassination of Archduke Franz Ferdinand of Austria-Hungary and drew in major powers through a web of alliances, resulting in unprecedented trench warfare and roughly 17 million deaths. World War II (1939-1945) began with Germany's invasion of Poland and became the deadliest conflict in history, involving most of the world's nations and resulting in an estimated 70 to 85 million deaths, including the systematic genocide of six million Jews during the Holocaust. The war ended in Europe with Germany's surrender in May 1945, and in the Pacific after the United States dropped atomic bombs on Hiroshima and Nagasaki in August 1945.

8.4 The Cold War

The Cold War (roughly 1947-1991) was a prolonged geopolitical standoff between the United States and the Soviet Union and their respective allies, characterized by ideological conflict between capitalism and communism, nuclear arms and space races, and proxy wars (such as in Korea, Vietnam, and Afghanistan) rather than direct military conflict between the two superpowers. It ended with the collapse of the Soviet Union in 1991, following events such as the fall of the Berlin Wall in 1989.

9. Economics & Finance

9.1 Supply and Demand

Supply and demand is a foundational economic model describing how the price of a good is determined in a market. Demand generally falls as price rises (the law of demand), while supply generally rises as price rises (the law of supply). The point at which the supply and demand curves intersect is the market equilibrium price, where the quantity buyers wish to purchase equals the quantity sellers wish to sell. Shifts in either curve, caused by factors like changes in income, consumer preferences, input costs, or technology, move the equilibrium price and quantity.

9.2 Inflation and Monetary Policy

Inflation is the general rise in prices over time, reducing the purchasing power of money; it is commonly measured using indices such as the Consumer Price Index (CPI). Central banks (such as the U.S. Federal Reserve) manage inflation primarily through monetary policy, most notably by adjusting interest rates: raising rates tends to cool spending and borrowing, reducing inflation, while lowering rates tends to stimulate economic activity. Excessive or runaway inflation (hyperinflation) can severely destabilize an economy, while deflation (falling prices) can discourage spending and investment.

9.3 GDP and Economic Growth

Gross Domestic Product (GDP) measures the total monetary value of all finished goods and services produced within a country in a given period, and is the most widely used indicator of a country's economic size and growth. GDP can be measured via the expenditure approach (consumption + investment + government spending + net exports), the income approach, or the production approach. A recession is commonly defined as two or more consecutive quarters of declining real GDP, while sustained GDP growth is generally associated with rising employment and living standards, though it does not by itself capture income inequality or environmental costs.

9.4 Stocks, Bonds, and Basic Investing Concepts

A stock (or share) represents partial ownership in a company, entitling the holder to a proportional claim on assets and earnings, and its price fluctuates based on company performance and investor sentiment. A bond is a debt instrument: the buyer lends money to an issuer (a government or corporation) in exchange for periodic interest payments and repayment of the principal at maturity. Diversification, spreading investments across different assets, is a core risk-management principle, since it reduces exposure to any single investment's poor performance.

10. Psychology & Cognitive Science

10.1 Major Psychological Perspectives

- **Behaviorism:** focuses on observable behavior shaped by conditioning (e.g. Pavlov's classical conditioning, Skinner's operant conditioning).
- **Cognitive psychology:** studies internal mental processes such as memory, perception, problem-solving, and language.
- **Psychoanalytic theory:** pioneered by Sigmund Freud, emphasizes unconscious drives and early childhood experiences.
- **Humanistic psychology:** emphasizes personal growth, free will, and self-actualization (e.g. Maslow, Rogers).
- **Biological psychology:** examines how brain structures, neurotransmitters, and genetics influence behavior.

10.2 Memory Systems

Human memory is commonly divided into sensory memory (very brief retention of sensory input), short-term/working memory (limited-capacity, active storage lasting seconds to minutes, used for reasoning and comprehension), and long-term memory (relatively permanent storage). Long-term memory is further split into explicit memory (consciously recalled facts and events, such as semantic and episodic memory) and implicit memory (unconscious, such as motor skills and conditioned responses). The process of moving information from short-term to long-term storage is called consolidation.

10.3 Cognitive Biases

Cognitive biases are systematic patterns of deviation from rational judgment. Confirmation bias is the tendency to favor information that confirms existing beliefs while discounting contradictory evidence. The anchoring effect describes over-reliance on the first piece of information encountered when making decisions. Availability heuristic refers to overestimating the likelihood of events that are more easily recalled, often because they are vivid or recent. The Dunning-Kruger effect describes a pattern in which people with limited knowledge or competence in a domain tend to overestimate their own ability.

10.4 Motivation and Emotion

Maslow's hierarchy of needs proposes that human motivation is organized in levels, from basic physiological needs (food, water, sleep) and safety, through social belonging and esteem, up to self-actualization (realizing one's full potential), with lower needs generally taking precedence over higher ones. Emotion is typically described as involving three components: a subjective feeling, a physiological response (such as changes in heart rate), and a behavioral or expressive response (such as facial expression). Intrinsic motivation arises from internal satisfaction in an activity itself, while extrinsic motivation is driven by external rewards or pressures.